

Used Car Price Prediction

Project Report

Key Findings from EDA

- `Present_Price` (the car's current market price) shows the strongest positive correlation with `Selling_Price` — it's the single best predictor of resale value.
- `Selling_Price` is right-skewed: most cars sell in the lower price range, with a smaller number of high-priced cars pulling the tail to the right.
- `Car_Age` (engineered from `Year`) shows a clear negative relationship with `Selling_Price` — older cars sell for less, confirming depreciation over time.
- Diesel cars and automatic-transmission cars tend to have a higher median `Selling_Price` than petrol/CNG and manual cars respectively.
- `Car_Name` (high cardinality) and `Owner` (very low variance — most cars have had 0 previous owners) carry little independent signal once the other features are included.

Features Used in the Final Model

- `Present_Price` — current market price of the car, in lakhs
- `Kms_Driven` — total kilometres driven
- `Car_Age` — engineered from `Year` (`Year` dropped after this step)
- `Fuel_Type` — Petrol, Diesel, or CNG (one-hot encoded)
- `Seller_Type` — Dealer or Individual (one-hot encoded)
- `Transmission` — Manual or Automatic (one-hot encoded)
- `Owner` — number of previous owners (0–3)

Car_Name was dropped as a low-signal, high-cardinality feature; Year was dropped after Car_Age was derived from it.

Final Model

Model: LightGBM (tuned)

Test R²: 0.9494

Test RMSE: 0.8162

LightGBM (tuned) achieved the highest Test R² and lowest Test RMSE among all 10 model configurations tried. Train R² (0.9810) vs. Test R² (0.9494) leaves a gap of about 0.032 — small enough to indicate reasonable generalization rather than severe overfitting. All tree-based models (Random Forest, XGBoost, LightGBM) clearly outperformed Linear Regression and Ridge (which topped out around 0.867 Test R²), suggesting the relationship between the features and `Selling_Price` involves non-linear effects and feature interactions beyond what the EDA's linear correlations alone captured.

Model Comparison

Sorted by Test R^2 descending:

Model	Train R^2	Test R^2	Test RMSE
LightGBM (tuned)	0.9810	0.9494	0.8162
Random Forest n=200	0.9932	0.9465	0.8391
Random Forest n=100	0.9929	0.9448	0.8523
XGBoost (tuned)	0.9988	0.9443	0.8559
LightGBM (default)	0.9773	0.9422	0.8724
XGBoost (default)	1.0000	0.9380	0.9033
Ridge $\alpha=1$	0.8190	0.8674	1.3213
Ridge $\alpha=0.1$	0.8197	0.8670	1.3232
Linear Regression	0.8197	0.8669	1.3237
Ridge $\alpha=10$	0.8140	0.8621	1.3471

Links

GitHub Repository: <https://github.com/rotoncsedu/Car-Price-Prediction>

Live Streamlit App: <https://rotoncsedu-car-price-prediction.streamlit.app/>

Submitted By

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